

Thesis 2.0 — Student Cloud Project Overview

MSc Cloud Computing · National College of Ireland

One page per student: services, scope, baseline, status, and deploy needs.

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13 student projects · single PDF

Notes: "Live AWS" means resources created in a real account. Simulator/moto/LocalStack results are not live cloud measurements. GCP and Azure are listed only where relevant; most artefacts are AWS-oriented or local.

1. Sri Venkat Bora (25164414)

Project: Multi-Threaded vs Distributed Matrix Scaling Workloads

Baseline paper

Sabir & Alebrahim (2025), Mathematics — coarse-grained LU / multi-threaded matrix methods. Published: 17 January 2025. DOI: 10.3390/math13020298.

Project in two lines

Compares multi-threaded and Dask-style distributed matrix workloads to measure scaling, CPU, and memory behaviour. Uses a local Python/Dask stack for reproducible runs rather than a multi-node cloud cluster.

AWS services

None required for the current artefact (local Dask / threading / multiprocessing).

Other services / tooling required

Python, NumPy, Dask (local cluster). Optional later: AWS EC2 or any VM fleet if true multi-node network effects are needed.

Cloud vendor scope

Not cloud-locked — local machine only today. Not GCP/Azure-specific.

Deploy?

No cloud deploy required for the submit-ready local study. Live EC2 would be optional validation only.

Project status

Submit-ready (local results). Honest STATUS: not run on live AWS EC2.

2. Mehak

Project: Cross-Head Fusion for Multi-Head Attention Cluster Telemetry Monitoring (MHSA-TDL)

Baseline paper

Thapliyal (2026), multi-head attention for SLA compliance monitoring. arXiv posted: 6 May 2026 (arXiv:2605.05354); IEEE ICDCS 2026 acceptance noted in project docs.

Project in two lines

Reproduces a per-metric multi-head attention monitor on synthetic cluster telemetry and tests a cross-head fusion fix for under-prediction in bursts. Training and evaluation run entirely offline on generated time series.

AWS services

None required.

Other services / tooling required

Python, PyTorch/scikit-learn style ML stack, synthetic telemetry generator, pytest.

Cloud vendor scope

Local / research ML only — not AWS, GCP, or Azure dependent.

Deploy?

No cloud deploy. Run training/eval locally.

Project status

Submit-ready (synthetic results, tests green, report PDF present).

3. Vishvaksen

Project: Hybrid IaC Misconfiguration Detection (rule-based + ML)

Baseline paper

War et al. (2025), semantics-aware IaC security-smell detection. arXiv:2509.18790 — first version published on arXiv: 23 September 2025.

Project in two lines

Builds a hybrid detector for Infrastructure-as-Code misconfigurations, combining rules with lightweight ML and testing the role of natural-language comments. Works on local IaC corpora and static analysis tooling.

AWS services

None required for core experiments (optional if scanning live CloudFormation later).

Other services / tooling required

Python, Checkov / policy scanners, scikit-learn, synthetic + real IaC samples.

Cloud vendor scope

Tooling is cloud-agnostic (Terraform/CloudFormation-style IaC). Not tied to GCP/Azure APIs.

Deploy?

No mandatory cloud deploy. Scan files locally.

Project status

Submit-ready (artefact + tests). PDF compile may need local LaTeX refresh.

4. Pooja

Project: Stability-Aware Predictive Kubernetes Autoscaling (PAKS)

Baseline paper

Wanigasooriya & Ekanayake (2026), NimbusGuard proactive Kubernetes autoscaling. arXiv:2604.11017 posted 13 April 2026; IEEE ICOIN 2026 DOI 10.1109/ICOIN68469.2026.11480646.

Project in two lines

Implements and compares reactive HPA-style, aggressive proactive, and stability-aware (smoothing + hysteresis) autoscaling controllers on synthetic workloads. Focuses on SLA vs scaling churn trade-offs.

AWS services

Optional — Amazon EKS only if you choose AWS as the Kubernetes host. Not required for the current synthetic evaluation.

Other services / tooling required

Kubernetes (local kind/minikube or any cluster), metrics-server / custom metrics, Python controllers, pytest.

Cloud vendor scope

Kubernetes-centric. Can run on local K8s, AWS EKS, GCP GKE, or Azure AKS — current pack is not locked to one vendor.

Deploy?

Local Kubernetes is enough for the artefact. Cloud K8s deploy is optional.

Project status

Submit-ready (synthetic evaluation, diagrams, report PDF).

5. Uday

Project: Federated QoS Offload Decisions for OneM2M IoT Middleware

Baseline paper

Et-Tousy, Zyane & Sharif (2026), adaptive QoS / ML for OneM2M. Journal DOI 10.1007/s10922-026-10071-4; Research Square preprint timeline includes November 2025 / May 2026 versions used for the open PDF.

Project in two lines

Reproduces a centralized Random Forest QoS offload classifier and extends it toward federated / multi-site decisions on synthetic OneM2M-style telemetry. Optional AWS Lambda packaging exists for edge-style inference.

AWS services

Optional: AWS Lambda (handler + SAM/template present) for deployed inference. Not required for local model experiments.

Other services / tooling required

Python, scikit-learn, federated/simulation utilities, pytest. OneM2M concepts (no live multi-site IoT required).

Cloud vendor scope

Primarily local ML. Optional AWS Lambda. Not GCP/Azure-specific.

Deploy?

Local runs sufficient. Live Lambda deploy optional for demo.

Project status

~98% submit-ready (pytest fixed; live AWS Lambda not deployed).

6. Nemi Ishwarlal Vikani (24303046)

Project: SecureFL-IDS — Privacy-Preserving Federated Intrusion Detection

Baseline paper

Saklani et al. (2026), privacy-preserving cloud-native IDS with federated learning + differential privacy. DOI 10.1109/iciss67859.2026.11454085 (IEEE ICISS 2026). Exact calendar day behind IEEE paywall; treat as 2026 conference publication.

Project in two lines

Trains a federated intrusion-detection model with privacy mechanisms on synthetic/local traffic features and documents gaps vs a full cloud-native deployment. Current evidence is from local simulation rounds.

AWS services

Designed for cloud-native contexts; live AWS (e.g., EKS/ECS + optional S3 datasets) not required for the local simulation pack.

Other services / tooling required

Python, Flower/FL stack or equivalent, differential privacy utilities, pytest.

Cloud vendor scope

Local simulation today. Conceptually AWS-oriented cloud-native IDS; not GCP/Azure-coded.

Deploy?

No live cloud deploy completed. Local training/eval only.

Project status

Marked complete for local artefact + report; baseline PDF still missing (IEEE). No live AWS.

7. Varun Gampa (23398639)

Project: Predictive Storage Cost Optimization Framework for Amazon S3

Baseline paper

Shen et al. (2025), TierBase (ICDE 2025). DOI 10.1109/ICDE65448.2025.00049. Open extended version on arXiv:2505.06556 (May 2025).

Project in two lines

Predicts S3 storage-class moves with rule-based (TierBase-inspired) and ML recommenders, estimating savings vs naive and lifecycle baselines. Runs in dry-run simulator by default; Boto3 path supports live S3 metadata.

AWS services

Amazon S3 (core). Optional: AWS Cost Explorer, CloudWatch, IAM for live metadata collection.

Other services / tooling required

Python, Boto3, XGBoost/Prophet, local workload simulator, pytest.

Cloud vendor scope

AWS-focused when live. Simulator mode needs no cloud. Not GCP/Azure object-store coded.

Deploy?

Not required for simulator results. Live S3 deploy/read of buckets optional (credentials + cost).

Project status

Submit-ready on simulator (DRY_RUN). Live AWS Cost Explorer / bucket campaigns not claimed.

8. Rasool Basha Durbesula (24205478)

Project: DynamoDB Partition-Key Design × Capacity Mode under Serverless Workloads

Baseline paper

Panteli■ et al. (2026), benchmarking SQL vs NoSQL persistence. Published: 15 January 2026 (Future Internet). DOI 10.3390/fi18010053.

Project in two lines

Factorial experiment on DynamoDB key designs and on-demand vs provisioned capacity under Zipfian serverless loads, measuring latency, throttles, and capacity cost. Terraform + Lambda generators included.

AWS services

Amazon DynamoDB, AWS Lambda (load generators), IAM, Amazon CloudWatch. Terraform for IaC.

Other services / tooling required

Python, Terraform, pytest/moto for local checks.

Cloud vendor scope

AWS-only for live study. Local moto can validate code without cloud.

Deploy?

Yes for real latency/capacity numbers: terraform apply + campaign. Moto alone is not live AWS.

Project status

Submit-ready artefact/report; evaluation evidence largely local/moto unless live campaign is run.

9. Kondragunta Lakshmi Chaitanya (25171216)

Project: Lambda Cold-Start Isolation across Runtimes and Deployment Packages

Baseline paper

Bluemke & Zdanowski (2025), evaluation of AWS Lambda configurations. Manuscript received 8 May 2025; journal issue IJET 71(3) 2025. DOI 10.24425/ijet.2025.153619.

Project in two lines

Measures cold-start and warm latency across languages/package shapes and isolation settings on AWS Lambda, comparing qualitative trends to the baseline study. IaC and test harness included.

AWS services

AWS Lambda (Python/Node/Java runtimes), IAM, Amazon CloudWatch Logs/Metrics. Packaging via zip/container as designed.

Other services / tooling required

Python driver, IaC (SAM/Terraform-style), pytest.

Cloud vendor scope

AWS-only for live cold-start truth. Local stubs do not replace INIT latency.

Deploy?

Yes for authoritative cold-start tables. Local validation is submit-ready but not live AWS.

Project status

Submit-ready (local validation). Live multi-runtime campaign still the main uplift.

10. Anjaneya Reddy Gurram (24288853)

Project: SQS Reliability and Recovery under Injected Consumer/Downstream Failures

Baseline paper

Kyrychenko, Ostapov & Kyrychenko (2025), optimization of SQS configurations for batch processing. Published: 12 March 2025 (WSEAS Transactions on Systems). DOI 10.37394/23202.2025.24.4.

Project in two lines

Studies SQS visibility timeout, redrive, and batch settings under controlled fault injection, with Lambda consumers and DynamoDB state. A local discrete-event simulator currently produces the published figures.

AWS services

Amazon SQS (+ DLQ), AWS Lambda, Amazon DynamoDB, AWS Systems Manager Parameter Store (fault switch), IAM, CloudWatch. SAM template present.

Other services / tooling required

Python simulator, SAM/laC, pytest/moto.

Cloud vendor scope

AWS for live campaigns. Simulator mode needs no cloud. Not GCP/Azure.

Deploy?

Yes for live fault campaigns. Simulator is enough for methodology demo; live backend unit-tested with moto only so far.

Project status

Submit-ready on simulator. Live AWS backend not executed against a real account.

11. Vikas Reddy Amanagantti (X25178849)

Project: Application-Level Idempotency Strategies for Lambda Retries + DynamoDB

Baseline paper

Qi et al., Halfmoon / asymmetric logging for stateful serverless. ACM TOCS DOI 10.1145/3725985 (2025 journal version); SOSP 2023 precursor DOI 10.1145/3600006.3613154 (conference publication year 2023).

Project in two lines

Compares three application-level write paths under injected retries, measuring duplicate mutations, latency, and DynamoDB capacity against the exactly-once idea from Halfmoon (without a custom runtime).

AWS services

AWS Lambda, Amazon DynamoDB (+ Streams), IAM, Amazon CloudWatch (alarms/metrics), Terraform. Region used in live work: eu-west-1.

Other services / tooling required

Python, Terraform, pytest/moto, analysis scripts.

Cloud vendor scope

AWS-only for live measurement. Moto validates correctness locally.

Deploy?

Yes. Infra can be applied with Terraform; pilot on live AWS completed (N=1000 chosen). Full campaign should use low concurrency (account limit ~10).

Project status

Artefact complete; live pilot done; full live campaign / report refresh still in progress vs moto-only STATUS text.

12. Yashaswini Penumarthy (24262404)

Project: Lightweight Fault Detection and Localisation in AWS Serverless Microservices

Baseline paper

Xing, Wang & Liu (2025), dual-channel deep fault localisation. Received 14 April 2025; accepted 27 May 2025; Sensors 25(11), 3396. DOI 10.3390/s25113396. (RCAEval benchmark also used: Pham et al. 2025, DOI 10.1145/3701716.3715290.)

Project in two lines

Implements a serverless order microservice with fault injection and a lightweight rule+ML detector, trading a little F1 for much lower monitoring overhead versus deep baselines on shared data.

AWS services

Amazon API Gateway, AWS Lambda (multi-function), Amazon DynamoDB, Amazon CloudWatch, AWS X-Ray, AWS Systems Manager Parameter Store, IAM. SAM template.

Other services / tooling required

Python, SAM/cfn-lint, pytest, RCAEval-style evaluation harness.

Cloud vendor scope

AWS-only for live fault campaigns. Simulation path exists for functional checks.

Deploy?

Yes for live calibration and fault campaigns (~budgeted AWS spend). Sim results are labeled non-AWS.

Project status

Submit-ready with simulated + RCAEval evidence. Live AWS campaigns not run.

13. Kasireddy Vadicharla (25104047)

Project: Serverless Log Anomaly Detection with Cross-System Transfer (ELFA-Log style)

Baseline paper

Zhao et al. (2025), ELFA-Log. Published: 10 July 2025 (Computers 14(7), 272). DOI 10.3390/computers14070272.

Project in two lines

Detects anomalies in serverless/CloudWatch-style logs, comparing source-labelled transfer (ELFA-Log-style) against weaker target-only one-class models. Prefer LocalStack for local AWS API emulation.

AWS services

Amazon CloudWatch Logs (target stream concept), optional Lambda for ingestion; LocalStack acceptable for local API parity. IAM as needed.

Other services / tooling required

Python, scikit-learn, Loghub public corpora, LocalStack, pytest.

Cloud vendor scope

AWS log APIs or LocalStack. Not GCP/Azure Logging coded as primary.

Deploy?

LocalStack/local first. Live CloudWatch optional for target-domain demo.

Project status

Baseline paper on disk; full completion of this project was previously skipped in the cohort finish pass — treat as not submit-ready unless separately completed.